Iman Gholami

Ph.D. Candidate in Computer Science

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Research Interests

Theoretical Computer Science: Approximation and online algorithms, combinatorial optimization, network design, graph algorithms, and algorithmic game theory.

AI and Formal Mathematics: Interactive theorem proving (Lean 4), autoformalization of research mathematics, and reinforcement learning for automated reasoning.

Education

2023 – 2028 (Expected Aug 2028)
University of Maryland, College Park | College Park, MD
Ph.D. Candidate in Computer Science
Advisor: Prof. MohammadTaghi Hajiaghayi

(Awarded May 2025)
M.Sc. in Computer Science
GPA: 3.97 / 4.00

2017 – 2022 (Awarded May 2022)
Sharif University of Technology | Tehran, Iran
B.Sc. in Computer Engineering
GPA: 3.57 / 4.00

Honors, Awards, & Fellowships

2025 **Best Paper Award**, IEEE Symposium on Foundations of Computer Science (FOCS)

2025 **MPower Early Scholars Award**, University of Maryland, College Park

2023 **2nd Place**, ACM ICPC North America South Division (Mid-Atlantic USA Region)

2023 **Dean's Fellowship**, University of Maryland, College Park

2021 **3rd Place**, Iranian National Scientific Olympiad in Computer Science and Engineering (undergraduate nationwide competition)

2020 **2nd Place**, ACM ICPC Asia-West Regional Contest, Tehran Site

2017 **Silver Medal**, International Olympiad in Informatics (IOI)

2017 **2nd Place**, ACM ICPC Asia-West Regional Contest, Tehran Site

2016 **Gold Medal**, Iran National Olympiad in Informatics (INOI)

2015 **Silver Medal**, Iran National Olympiad in Informatics (INOI)

PUBLICATIONS

Full publication list on Google Scholar

Authors are listed alphabetically unless otherwise indicated.

Refereed Journal Articles

2-Approximation for Prize-Collecting Steiner Forest — **JACM** 2025
Ali Ahmadi, **Iman Gholami**, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, and Mohammad Mahdavi.
Journal of the ACM, Vol. 72, No. 2, Article 17, pp. 1–27.

Refereed Conference Proceedings

Quiet Planting for k-SAT, Multiple Solutions of Arbitrary Geometry — **COLT** 2026
Ali Ahmadi, Kiarash Banihashem, **Iman Gholami**, MohammadTaghi Hajiaghayi, and Jan Olkowski.
39th Conference on Learning Theory, PMLR Vol. 336, pp. 75–105.

Bi-Criteria Metric Distortion — **ICLR** 2026
Kiarash Banihashem, Diptarka Chakraborty, Shayan Chashm Jahan, **Iman Gholami**, MohammadTaghi Hajiaghayi, Mohammad Mahdavi, and Max Springer.
14th International Conference on Learning Representations.

Breaking a Long-Standing Barrier: $2 - \varepsilon$ Approximation for Steiner Forest — **FOCS** 2025
Ali Ahmadi, **Iman Gholami**, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, and Mohammad Mahdavi.
Best Paper Award at the 66th IEEE Symposium on Foundations of Computer Science.
(Journal version: under revision, Journal of the ACM.)

Prize-Collecting Forest with Submodular Penalties: Improved Approximation — **IPCO** 2025
Ali Ahmadi, **Iman Gholami**, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, and Mohammad Mahdavi.
26th International Conference on Integer Programming and Combinatorial Optimization, LNCS.
(Journal version: revised manuscript under review, Operations Research.)

Prize-Collecting Steiner Tree: A 1.79 Approximation — **STOC** 2024
Ali Ahmadi, **Iman Gholami**, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, and Mohammad Mahdavi.
56th ACM Symposium on Theory of Computing, pp. 1641–1652.
(Journal version: revised manuscript under review, SIAM Journal on Computing.)

2-Approximation for Prize-Collecting Steiner Forest — **SODA** 2024
Ali Ahmadi, **Iman Gholami**, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, and Mohammad Mahdavi.
35th ACM-SIAM Symposium on Discrete Algorithms, pp. 669–693.

Preprints & Working Papers

Nonuniform Buy-at-Bulk Network Design: Improved Approximation and LP Integrality Gap — **arXiv** 2026
Iman Gholami and MohammadTaghi Hajiaghayi.
arXiv:2611.06894, November 2026.
Lean 4 formalization: github.com/igholami/buy-at-bulk-lean

The Tight Competitive Ratio of Greedy for Online Steiner Forest — **arXiv** 2026
Iman Gholami, Ilya Hajiaghayi, MohammadTaghi Hajiaghayi, and Peyman Jabbarzade.
arXiv:2610.40569, October 2026.
Lean 4 formalization: github.com/igholami/online-steiner-forest-lean

Constructive Arboricity in Almost-Linear Time for Dense Graphs — **arXiv** 2026
Iman Gholami, MohammadTaghi Hajiaghayi, Mahdi JafariRaviz, Alireza Kaviani, Soheil Mohammadkhani, and AmirMohammad Shahrezaei.
arXiv:2610.13937, October 2026. (Under review at STOC 2027.)

Beyond the Library: An Agentic Framework for Autoformalizing Research Mathematics — **arXiv** 2026
Arshia Soltani Moakhar, **Iman Gholami**, Max Springer, Mahdi JafariRaviz, and MohammadTaghi Hajiaghayi (contribution order).
arXiv:2606.31134, June 2026. (Under review at ICLR 2027.)

Talks & Presentations

July 2026 — **Quiet Planting for k-SAT, Multiple Solutions of Arbitrary Geometry**
Conference on Learning Theory (COLT 2026), San Diego, CA

May 2025 — **Breaking a Long-Standing Barrier: $2 - \varepsilon$ Approximation for Steiner Forest**
Capital Area Theory Seminar (CATS), University of Maryland, College Park

Sept 2023 — **2-Approximation for Prize-Collecting Steiner Forest**
Capital Area Theory Seminar (CATS), University of Maryland, College Park

Research & Industry Appointments

2023 – Present — **Graduate Research Assistant**
University of Maryland, College Park

- Co-developed an approximation algorithm breaking the longstanding factor-2 barrier for Steiner Forest, recognized with a FOCS 2025 Best Paper Award.
- Developed improved approximation guarantees and studied LP integrality gaps for fundamental network design and prize-collecting optimization problems.
- **DARPA expMath (Exponentiating Mathematics):** Formalized research-level algorithmic results in Lean 4 and investigated AI-assisted theorem proving, including approaches to formalizing mathematical proofs.
- **DARPA QuICC (Quantum-Inspired Classical Computing):** Investigating computational hardness and quantum-inspired approaches to optimization.
- Investigating algorithmic game theory and social choice, focusing on the design and analysis of algorithms for collective decision-making.

Jun – Aug 2026 — **Software Engineer Intern**
Intrinsic Technologies (Axiom), New York, NY

- Contributed to formal verification of zero-knowledge virtual machines (ZK VMs) in Lean 4.
- Designed and implemented a reinforcement learning (RL) evaluation infrastructure to optimize Large Language Models (LLMs) and reasoning models for automated theorem proving.

May – Aug 2025 — **Software Engineer Intern**
Axal (YC W25), San Francisco, CA

- Architected and built a platform combining static analysis, code dependency graphs, and LLM reasoning from the ground up to automate modernization of large-scale legacy systems.

Nov 2018 – Dec 2022 — **Software Engineer & Team Leader**
Balad (Navigation & Maps Platform, 30M+ Installs), Tehran

- Architected a real-time traffic prediction engine that improved ETA accuracy by 35%.
- Refactored the core routing and navigation service from Java to Kotlin, reducing system latency by 30%.
- Redesigned the analytics data pipeline (ClickHouse, Airflow, Apache Spark), cutting compute costs by 30%.

Jul – Oct 2018 — **Software Engineer Intern**
Bazaar (Application Marketplace, 100M+ Installs), Tehran

- Built the internal ledger and wallet service backing in-app transactions for 50M+ users and resolved a critical concurrency bug that had frozen $20,000 in user funds.

Teaching Experience

University of Maryland, College Park — Graduate Teaching Assistant

Spring 2026	CMSC 423: Computational Genomics (Prof. Rob Patro)
Fall 2025	CMSC 474: Introduction to Computational Game Theory (Prof. MohammadTaghi Hajiaghayi)
Fall 2024	CMSC 414: Computer and Network Security (Michael Marsh)
Fall 2023	CMSC 474: Introduction to Computational Game Theory (Prof. MohammadTaghi Hajiaghayi)
Spring 2023	CMSC 351: Algorithms (Max Morowski & Prof. Clyde Kruskal)

Sharif University of Technology — Undergraduate Teaching Assistant

Fall 2021	Theory of Machines and Languages (Prof. Ali Movaghar)
Fall 2021	Design of Algorithms (Prof. Hamid Zarrabi-Zadeh)
Spring 2021	Modern Information Retrieval (Prof. Mahdieh Soleymani Baghshah)
Spring 2021	Discrete Structures (Prof. Hamid Zarrabi-Zadeh)
Fall 2019	Design of Algorithms (Prof. Mohammad Ali Abam)

Professional Service

Conferences — **External Reviewer:**
IEEE Symposium on Foundations of Computer Science (FOCS 2026), ACM Symposium on Theory of Computing (STOC 2026), ACM-SIAM Symposium on Discrete Algorithms (SODA 2026, 2027), International Colloquium on Automata, Languages, and Programming (ICALP 2025), ACM Symposium on Parallelism in Algorithms and Architectures (SPAA 2025), IEEE Conference on Decision and Control (CDC 2026).

Journals — **Referee:**
Pacific Journal of Optimization.

Technical Skills

Languages	C++, Python, Lean 4 (Interactive Theorem Proving), Kotlin, TypeScript, Go, Java, SQL, Bash
Theory & Math	Approximation & Online Algorithms, Linear/Semidefinite Programming, Primal-Dual Methods, Submodular Optimization, Graph Theory, Mechanism Design, Formal Verification
AI & Systems	Automated Theorem Proving, Reinforcement Learning for Reasoning Models, Distributed Systems, ClickHouse, Apache Spark, Airflow, Docker, Kubernetes, PostgreSQL, Redis, Linux

References

Ph.D. Advisor — **Prof. MohammadTaghi Hajiaghayi**
Jack and Rita G. Minker Professor of Computer Science
Department of Computer Science, University of Maryland, College Park
Email: hajiagha@umd.edu

Collaborator — **Dr. Diptarka Chakraborty**
Assistant Professor, School of Computing
National University of Singapore
Email: diptarka@comp.nus.edu.sg